

## Quick\_Activity\_2.pdf

A canteen sells rice meals at ₱85 each and drinks at ₱25 each. Every order receives a 10% discount on the rice-meal subtotal only. The canteen also charges a ₱5 packaging fee per rice meal.

### Problem Analysis

| Inputs                                                       | Processes                                                                                                                                                                                                                                          | Outputs                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Number of rice meals,<br>number of drinks, payment<br>amount | Calculate rice meals<br>Subtotal, calculate the 10%<br>of rice meal Subtotal,<br>subtract the discount,<br>calculate drink total, add<br>the packaging fee per rice<br>meal, calculate change by<br>subtracting the final bill<br>from the payment | Meal subtotal<br>Drink subtotal<br>Discount<br>Packaging fee<br>Final bill<br>Change |

### Decomposition

1. Input the number of rice meals.
2. Input the number of drinks.
3. Input the payment.
4. Calculate the meal subtotal.
5. Calculate the 10% discount on the meal subtotal.
6. Calculate the drink cost.
7. Calculate the packaging fee.
8. Calculate the final bill.
9. Calculate the change.
10. Display all monetary results to two decimal places.

## Pseudocode

START

```
    READ ricemealSubtotal drinkCost, discountedSubtotal,  
    packagingFeePerMeal, payment  
    ricemealSubtotal  $\leftarrow$  quantity  $\times$  85  
    discount  $\leftarrow$  ricemealSubtotal  $\times$  10%  
    discountedSubtotal  $\leftarrow$  ricemealSubtotal - discount  
    drinkCost  $\leftarrow$  quantity  $\times$  25  
    packagingFeePerMeal  $\leftarrow$  quantity  $\times$  5  
    totalCost  $\leftarrow$  discountedSubtotal + drinkCost + packagingFeePerMeal  
    change  $\leftarrow$  payment - totalCost  
    DISPLAY totalCost, change
```

END

## Dry-Run Calculation

### Test Case 1

Input:

- Rice meal: 3
- Drink: 4
- Payment: ₱500.00

| Step | Variable            | Calculation                                       | Result  |
|------|---------------------|---------------------------------------------------|---------|
| 1    | ricemealSubtotal    | $3 \times \text{₱}85.00$                          | ₱255.00 |
| 2    | discount            | $0.10 \times \text{₱}255.00$                      | ₱25.50  |
| 3    | discountedSubtotal  | $\text{₱}255 - \text{₱}25.50$                     | ₱229.50 |
| 4    | drinkCost           | $4 \times \text{₱}25.00$                          | ₱100.00 |
| 5    | packagingFeePerMeal | $3 \times \text{₱}5.00$                           | ₱15.00  |
| 6    | totalCost           | $\text{₱}229.50 + \text{₱}100.00 + \text{₱}15.00$ | ₱344.50 |
| 7    | change              | $\text{₱}500.00 - \text{₱}344.50$                 | ₱155.50 |

## Test Case 2

Input:

- Rice meal: 5
- Drink: 3
- Payment: ₱600

| Step | Variable            | Calculation                                      | Result  |
|------|---------------------|--------------------------------------------------|---------|
| 1    | ricemealSubtotal    | $5 \times \text{₱}85.00$                         | ₱425.00 |
| 2    | discount            | $0.10 \times \text{₱}425.00$                     | ₱42.50  |
| 3    | discountedSubtotal  | $\text{₱}425.00 - \text{₱}42.50$                 | ₱382.50 |
| 4    | drinkCost           | $3 \times \text{₱}25.00$                         | ₱75.00  |
| 5    | packagingFeePerMeal | $5 \times \text{₱}5.00$                          | ₱25.00  |
| 6    | totalCost           | $\text{₱}382.50 + \text{₱}75.00 + \text{₱}25.00$ | ₱482.50 |
| 7    | change              | $\text{₱}600.00 - \text{₱}482.50$                | ₱117.50 |

**Explain:** Why would applying the 10% discount to the entire bill produce an incorrect result?

Applying the 10% discount to the entire bill would be incorrect because it states that the discount applies only to the rice-meal subtotal. The drinks and packaging fees do not receive a discount. Therefore, they must be added at their full prices after the rice-meal discount is calculated.